7/5/2016(M)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
April - May 2016

Max. Marks: 100 | **Duration: 3 hrs**
Class: SE | Semester: IV
Name of the Course: Relational Database Management System | Branch: Computer
Course Code: UCEC4O4

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Draw ER model for online Railway Ticket Reservation system. Convert ER diagram into tables.	10 M
Q 1 (b)	Give the advantages of DBMS over file systems.	10 M
Q2 (a)	Explain the following relational algebra operations: (i) Natural Join (ii) Assignment (iii) Generalised Projection (iv) Select	10 M
Q2 (b)	For the following given database, write SQL queries :- person(driver_id #, name, address) car(license, model, year) accident(report _no, date, location) owns(driver_id #, license) participated(driver_id, car, report _number, damage_ amount) i. Find the total number of people who owned cars that were involved in accident in 1995. ii. Find the number of accidents in which the cars belonging to "Sunil K." were involved. iii. Update the damage amount for car with license number "Mum2022" in the accident with report number "AR2197" to Rs. 5000.	10 M
Q3 (a)	What are triggers? Give an example. Ilustrate the cases when triggers must not be used.	10 M

Q3 (b) Q3(b)	What is a view? How it is defined and stored? What are the benefits and limitations of views ? OR What is Security and authorization mechanism in database.	10 M 10 M
Q4 (a) Q4(a)	Explain types of constraints with example. **OR** Explain referential integrity with example.	10 M 10M
Q4 (b)	Explain 2NF,3NF and BCNF with suitable example	10 M
Q5 (a) Q5(a)	When Transaction is rolled back under timestamp ordering, it is assigned a new timestamp? Why can it not simply keep its old timestamp. **OR** Explain in detail Log based Recovery	10 M 10 M
Q5 (b)	Write short notes on: **Any 2** i) Shadow Paging ii) Database Architecture iii) Assertions iv) Data Independence	10 M